Computer Science - K-12 Level Assessment 25 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following has the greatest potential for compromising a user's personal privacy?
   (a) A group of cookies stored by the user's Web browser
   (b) The Internet Protocol (IP) address of the user's computer
   (c) The user's e-mail address
   (d) The user's public key used for encryption

2. Let l = [1,2,3,4]. What is max(l) in Python 3?
   (a) 1
   (b) 2
   (c) 3
   (d) 4

3. In Python 3, let r = lambda q: q * 2. What is r(3)?
   (a) 2
   (b) 6
   (c) 3
   (d) 1

4. In python 3, which of the following is floor division?
   (a) /
   (b) //
   (c) %
   (d) |

5. Which is the smallest asymptotically?
   (a) O(1)
   (b) O(n)
   (c) O(n^2)
   (d) O(log n)

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression ['Hi!'] * 4 produces the list ['Hi!', 'Hi!', 'Hi!', 'Hi!'].
7. True or False: In Python 3, accessing the first element of the tuple ('abcd', 786, 2.23, 'john', 70.2) returns 'abcd'.
8. True or False: The boolean expression a[i] == max || !(max != a[i]) simplifies to a[i] != max.
9. True or False: The program may have bugs even if extensive testing found no errors.
10. True or False: Python variable names are not case-sensitive and can be used interchangeably regardless of capitalization.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the first duplicate element in a given array of integers.
12. Write a function to find three closest elements from three sorted arrays.

End of Paper